

ECON 21030: Honors Econometrics

Midterm 1

Spring Quarter 2024

There are a total of **120 points** on the exam. I recommend you spend about 20 minutes each on the first three problems and 30 minutes on the final problem. Use the remaining time to check your work and revisit parts you skipped.

1: [30 points] True, false, or it depends? If true, prove it or give an appropriate reference (briefly). If false, explain why or provide a counterexample. If it depends, explain why, preferably with an example. No credit will be given without some justification, and good justification will be awarded full credit.

1. [5 points] If for random variables Y and X it happens that $E[Y|X] = E[Y]$, then $E[X|Y] = E[X]$.

This false. While full independence of Y and X would deliver this result, mean independence alone does not. For example, suppose that $Y = XU$, where $X \sim \text{bernoulli}(0.5)$ and $U+0.5 \sim \text{bernoulli}(0.5)$, with X and U independent. Then $E[Y|X] = E[XU|X] = 0$. But $E[X|Y]$ is 1 if $Y \neq 0$, and 0 otherwise.

2. [5 points] For random vector X with mean μ and variance-covariance Σ , $AX \sim N(A\mu, A\Sigma A')$ for known matrix A .

This is false. While the variance of AX is indeed $A\Sigma A'$, only linear transformations of normals are normal (i.e., X would need to be normal too).

3. [5 points] A p-value indicates the likelihood that the null hypothesis is true.

False. P-values indicate the probability of type-1 error from a testing procedure that rejects the null when the evidence is at least as strong as the case that generated the p-value.

4. [5 points] Consider a regression of the random variable Y_i onto a constant and another random variable X_i , i.e. $Y_i = \alpha + \beta X_i + U_i$. Adding another covariate W_i to this regression will not affect β if and only if $\text{Cov}(Y_i, W_i) = 0$.

Also false. This covariance can be non-zero and β will still be unaffected if $\text{Cov}(X_i, W_i) = 0$.

5. [5 points] Under homoscedasticity, the sampling variance-covariance of the OLS estimator $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ is always diagonal.

False. Consider the example from lecture where the slope coefficient is a difference in means.

2: [25 points] Suppose we have an iid sample $\{X_i\}_{i=1}^N$. Let $\mu = E[X_i]$, $\sigma^2 = \text{Var}(X_i)$, and $\hat{\mu} = \frac{1}{N} \sum_{i=1}^N X_i$.

1. [5 points]: What is the asymptotic distribution of $\sqrt{N}(\hat{\mu} - \mu)$?

This is normal with mean zero and variance σ^2 .

2. [5 points]: Assume that σ^2 is known, use this asymptotic approximation to test the null hypothesis $H_0 : \mu = 0$. Write down the test statistic and the reference distribution.

The approximation says that under the null $\sqrt{N}\hat{\mu} \sim N(0, \sigma^2)$. So $\hat{\mu} \sim N(0, \sigma^2/N)$.

We can therefore use a boring old z-score as $z = \hat{\mu}\sqrt{N}/\sigma$ and compare to the standard normal distribution.

3. [5 points]: Now consider $\gamma = \exp(\mu)$ and $\hat{\gamma} = \exp(\hat{\mu})$. What is the asymptotic distribution of $\hat{\gamma}$?

By the delta method, we know that $\sqrt{N}(\hat{\gamma} - \gamma) \rightarrow^D N(0, \gamma^2 \sigma^2)$.

4. [5 points]: Use this approximation to test the null hypothesis $\tilde{H}_0 : \gamma = 1$. Write down the test statistic and the reference distribution.

Here we would use $\tilde{z} = \frac{(\exp(\hat{\mu})-1)\sqrt{N}}{\sigma \exp(\hat{\mu})}$

5. [5 points]: Is it possible for H_0 to be false but $\tilde{H}_0$ to be true or vice versa? Is it possible to reject your test of one by not the other? How do you explain this fact?

Clearly if $\mu = 0$ then $\gamma = \exp(\mu) = 1$, so the two nulls imply each other. Clearly also if $\hat{\mu} = 0$ we would have the same test statistic. But in general, $\tilde{z}$ need not equal z . If $\sigma = 1$, for example, then the difference in the test statistics will be:

$$z - \tilde{z} = \sqrt{N}(\hat{\mu} + \exp(-\hat{\mu}) - 1)$$

which you can show to be equal to zero at $\hat{\mu} = 0$ and have derivative proportional to $1 - \exp(-\hat{\mu})$, so generally z will be bigger than $\tilde{z}$!

What's going on here is that the delta method is providing a second layer of approximation that is only good when $\hat{\mu}$ gets close to μ . The invariance of these types of Z-tests (which are like Wald tests of a single restriction) to different parameterizations of the null is a motivation for other tests you will study in more advanced courses.

3: [25 points] Consider an iid sample $\{(Y_i, X_i)\}_{i=1}^N$ from a joint distribution with finite moments.

1. [5 points]: Prove that the sample average $\hat{\mu} = \frac{1}{N} \sum_{i=1}^N Y_i X_i$ is consistent for $E[Y_i X_i]$.

This is a straightforward application of the WLLN, which will apply when $\text{Var}(Y_i X_i) < \infty$.

2. [5 points]: Define $\hat{\gamma} = \left(\frac{1}{N} \sum_{i=1}^N X_i\right) \left(\frac{1}{N} \sum_{i=1}^N Y_i\right)$. Derive probability limit of $\hat{\gamma}$. Under what conditions does $\hat{\gamma} \rightarrow^P E[Y_i X_i]$?

This converges (by Slutsky) to $E[X_i]E[Y_i]$. This will be equal to $E[Y_i X_i]$ iff $\text{Cov}(Y_i, X_i) = 0$.

3. [15 points]: Is $\hat{\gamma}$ unbiased for its probability limit? Evaluate its expectation to show why or why not, and discuss how bias is affected by the conditions you established in the previous sub-question. *Hint: You may want to use the fact that $\hat{\gamma}$ can be expressed as $\frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N X_i Y_j$.*

While it is true that $E\left[\frac{1}{N} \sum_{i=1}^N X_i\right] = E[X_i]$ and $E\left[\frac{1}{N} \sum_{i=1}^N Y_i\right] = E[Y_i]$, as suggested by the hint, we have:

$$\begin{aligned} E[\hat{\gamma}] &= \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N E[X_i Y_j] \\ &= \frac{1}{N^2} \left(\sum_{i=1}^N E[X_i Y_i] + \sum_{i=1}^N \sum_{j \neq i}^N E[X_i Y_j] \right) \end{aligned}$$

Independence implies that $E[X_i Y_j] = E[X_i]E[Y_j] = E[X_i]E[Y_i]$ for $i \neq j$. But as you (should) have argued in the previous sub-problem, $E[X_i Y_i] = Cov(Y_i, X_i) + E[X_i]E[Y_i]$. Thus:

$$\begin{aligned} E[\hat{\gamma}] &= \frac{1}{N^2} \left(\sum_{i=1}^N [Cov(Y_i, X_i) + E[X_i]E[Y_i]] + \sum_{i=1}^N \sum_{j \neq i}^N E[X_i]E[Y_i] \right) \\ &= E[X_i]E[Y_i] + \frac{1}{N} Cov(Y_i, X_i) \end{aligned}$$

Bias comes from the fact that the same observations are used to compute the sample means of Y_i and X_i . Of course if $Cov(Y_i, X_i) = 0$, then there is no bias from doing so.

- 4: [40 points] Suppose you are studying the log earnings of U Chicago graduates at age 30, which we will denote as the random variable Y_i . You are curious how earnings vary with the students primary major (e.g., economics, art history, etc.). Let $X_i = (X_{i1}, \dots, X_{ik})'$ denote a k -dimensional vector of mutually exclusive indicators for each student's major choice. That is $\sum_{l=1}^k X_{il} = 1 \forall i$ and $X_{il} \in \{0, 1\} \forall l \in \{1, \dots, k\}$.

You have access to an iid sample $\{(Y_i, X_i')\}_{i=1}^N$ and estimate the following regression:

$$Y_i = X_i' \beta + \epsilon_i$$

1. [5 points]: Describe how to interpret β and provide a closed form expression for each entry.

We have $\beta_l = \frac{\sum_{i=1}^N X_{il} Y_i}{\sum_{i=1}^N X_{il}}$, i.e., the sample average Y_i among individuals with $X_{il} = 1$.

You can see this from the fact the design matrix $\mathbf{X}'\mathbf{X}$ is a k -by- k diagonal matrix with typical entry $(\mathbf{X}'\mathbf{X})_{ll} = \sum_{i=1}^N X_{il}$. Likewise $\mathbf{X}'\mathbf{Y}$ is a k -by-1 vector with typical entry $(\mathbf{X}'\mathbf{Y})_l = \sum_{i=1}^N X_{il} Y_i$.

2. [5 points]: Now we would like to study how earnings vary by students' participation in campus Greek life. Let G_i be a binary random variable that equals one if student i was a

member of a fraternity or sorority. The regression of interest is now:

$$Y_i = \alpha_0 G_i + X_i' \beta + \epsilon_i$$

Describe how you would interpret estimates of α_0 .

Here I am looking for something like differences in log earnings after adjusting for differences in major choice.

3. [5 points]: A friend tells you that when the number of majors (i.e., k) is very big, it might be hard to compute estimates of α_0 . Explain why this would be the case.

Inverting the design matrix could be computationally challenging if it is big.

4. [10 points]: Your friend instead suggests you estimate this regression:

$$Y_i = \alpha_1 G_i + \alpha_2 \bar{G}_i + \eta_i$$

where $\bar{G}_i$ is the average of G_i among students with student i 's major m , i.e., $E[G_i|X_i]$.

What is the relationship between α_1 and α_0 ?

They are the same, which you can see from applying FWL. In both cases, $\tilde{G}_i = G_i - \bar{G}_i$. This happens in the second case (with $\bar{G}_i$ as the regressor) because a regression of G_i on $\bar{G}_i$ yields a slope of one because:

$$\begin{aligned} \frac{E[G_i \bar{G}_i]}{E[\bar{G}_i^2]} &= \frac{E[(\bar{G}_i + U_i) \bar{G}_i]}{E[\bar{G}_i^2]} \\ &= \frac{E[\bar{G}_i \bar{G}_i]}{E[\bar{G}_i^2]} = 1 \end{aligned}$$

where the last line follows from the fact that $U_i = G_i - E[G_i|X_i]$, and hence is uncorrelated with $\bar{G}_i = E[G_i|X_i]$. The same holds exactly in sample.

5. [5 points]: Another friend tells you you're both wrong: You shouldn't be running one regression, you should be running many! Describe how to interpret the coefficient γ_m in the regression:

$$Y_i = \gamma_0 + \gamma_m G_i + U_i$$

if we have *only* included individuals with $X_{im} = 1$. Show that the we can compute the γ_m for each value in the support of X_i (i.e., each X_{im}) as: $\gamma_m = Cov(Y_i, G_i - \bar{G}_i | X_i) / Var(G_i - \bar{G}_i | X_i)$ (hint: use Frisch-Waugh to absorb the constant!).

This is just the difference in mean Y_i for students in major m , so $E[Y_i | G_i = 1, X_{im} = 1] - E[Y_i | G_i = 0, X_{im} = 1]$.

The second part follows from the hint. If we think of this regression as a regression of Y_i on a constant and G_i , the FWL tells us that γ_m can be computed as stated.

6. [10 points]: The same friend says you should be most interested in the simple average of γ_m s, or $\gamma = \frac{1}{k} \sum_{m=1}^k \gamma_m$. Will γ be the same as either α_0 or α_1 ? Why or why not? (hint: use the laws of total covariance / variance to interpret α_0 or α_1)

In general no. We know from our FWL tricks earlier that:

$$\alpha_0 = \alpha_1 = \frac{Cov(Y_i, G_i - \bar{G}_i)}{Var(G_i - \bar{G}_i)}$$

The numerator can be expressed using the law of total covariance as:

$$\begin{aligned} Cov(Y_i, G_i - \bar{G}_i) &= E[Cov(Y_i, G_i - \bar{G}_i | X_i)] + Cov(E[Y_i | X_i], E[G_i - \bar{G}_i | X_i]) \\ &= E[Cov(Y_i, G_i - \bar{G}_i | X_i)] \\ &= E[\gamma_{m(i)} Var(G_i - \bar{G}_i | X_i)] \end{aligned}$$

where the $m(i)$ notation means we want the γ_m for i 's major.

Likewise, the denominator is simply

$$Var(G_i - \bar{G}_i) = E[Var(G_i - \bar{G}_i | X_i)]$$

Thus:

$$\alpha_0 = \alpha_1 = E \left[\frac{Var(G_i - \bar{G}_i | X_i)}{E[Var(G_i - \bar{G}_i | X_i)]} \gamma_{m(i)} \right]$$

which is a different weighted average of the γ_m s.